

AE370 Project 2: Reentry-Coupled Transient Heat Conduction in a Spherical Thermal Protection System

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This project studies transient, radially symmetric heat conduction in a spherical vehicle subjected to reentry-like aerodynamic heating. A simplified point-mass trajectory with an exponential atmosphere provides altitude and velocity histories, which are mapped to an inward surface heat flux using the Sutton–Graves correlation. The resulting forced spherical heat equation is discretized in space by a conservative finite-volume method on concentric shells and advanced in time with a second-order Crank–Nicolson scheme, yielding a tridiagonal solve each step. Solver correctness is demonstrated through verification: (i) a method of manufactured solutions confirming second-order convergence in space and time, and (ii) an analytical cross-check for a homogeneous sphere under constant inward heat flux. Discretization studies are then used to select $(N, \Delta t_{\text{cond}}, \Delta t_{\text{traj}})$ so numerical errors are negligible for key metrics such as surface and TPS back-face temperatures.

I. Nomenclature

r	=	radial coordinate (m)
R	=	sphere radius (m)
T	=	temperature (K)
T_0	=	initial uniform temperature (K)
ρ	=	density (kg/m ³)
c_p	=	specific heat (J/(kg·K))
k	=	thermal conductivity (W/(m·K))
α	=	thermal diffusivity, $\alpha = k/(\rho c_p)$ (m ² /s)
N	=	number of finite-volume cells
Δr	=	radial grid spacing, $\Delta r = R/N$ (m)
Δt_{cond}	=	conduction time step (s)
Δt_{traj}	=	trajectory/forcing sampling time step (s)
$q_{\text{in}}(t)$	=	inward surface heat flux into the solid (W/m ²)
$A(r)$	=	spherical area, $A(r) = 4\pi r^2$ (m ²)
V_i	=	control-volume (shell) volume (m ³)
R_n	=	nose (stagnation) radius used in Sutton–Graves (m)
K_{SG}	=	Sutton–Graves correlation constant

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$h(t)$	=	altitude along trajectory (m)
$V(t)$	=	speed magnitude along trajectory (m/s)
m	=	vehicle mass (kg)
C_d	=	drag coefficient
A_{ref}	=	reference area for drag (m ²)
δ_{TPS}	=	TPS thickness (m)
r_{back}	=	TPS back-face radius, $r_{\text{back}} = R - \delta_{\text{TPS}}$ (m)

II. Introduction

The heat equation is one of the most important differential equations in engineering and appears in many contexts where diffusion processes govern system behavior. This project examines a simplified but physically meaningful initial boundary value problem (IBVP) for transient heat conduction in a spherical vehicle during atmospheric reentry. During reentry, large aerodynamic heat fluxes act at the surface and diffuse inward through a thermal protection system (TPS), making temperature predictions at the surface and at the TPS back-face important for survivability and design.

Because full reentry aerothermodynamics and material response are highly complex, we employ a reduced-order model: (i) a simple trajectory model provides altitude and speed histories, (ii) a Sutton–Graves correlation maps those histories to a time-dependent surface heat-flux forcing, and (iii) a one-dimensional radially symmetric conduction model predicts the internal temperature field. Prior investigations by Park *et al.* support the use of the Sutton–Graves correlation as a viable and accurate model [1]. The goal is to implement and verify a robust numerical method for this coupled forcing–conduction problem, then use it to explore how material properties influence surface and internal heating trends.

In contrast to a standalone heat-equation IBVP with prescribed boundary forcing, this project emphasizes three extensions: (i) a coupled forcing model in which $q_{\text{in}}(t)$ is generated from an explicit trajectory and atmosphere model rather than assumed a priori, (ii) multi-layer, discontinuous material properties handled in conservative form with face-consistent conductivities, and (iii) separate discretizations for the forcing sampling and the conduction solve ($\Delta t_{\text{traj}}, \Delta t_{\text{cond}}$), with sensitivity studies used to ensure reported trends are not numerical artifacts.

III. Dynamical System

Description of dynamical system

We model transient heat conduction in a radially symmetric spherical body with piecewise-constant material properties (TPS layer + inner structure). The governing equation is written in conservative spherical form:

$$\rho(r)c_p(r) \frac{\partial T}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(k(r) r^2 \frac{\partial T}{\partial r} \right), \quad 0 < r < R, \quad t > 0. \quad (1)$$

The initial condition is a uniform baseline temperature,

$$T(r, 0) = T_0, \quad 0 \leq r \leq R. \quad (2)$$

At the center, symmetry enforces zero radial gradient,

$$\left. \frac{\partial T}{\partial r} \right|_{r=0} = 0. \quad (3)$$

At the outer surface $r = R$, we impose an *inward* heat flux $q_{\text{in}}(t)$ into the solid. With r increasing outward, the inward flux convention corresponds to:

$$k(R) \left. \frac{\partial T}{\partial r} \right|_{r=R} = q_{\text{in}}(t), \quad q_{\text{in}}(t) > 0 \Rightarrow \text{heat enters the solid.} \quad (4)$$

The reentry heating flux is modeled by the Sutton–Graves correlation,

$$q_{\text{in}}(t) = K_{\text{SG}} \frac{\sqrt{\rho_{\infty}(h(t))}}{\sqrt{R_n}} |V(t)|^3, \quad (5)$$

where $h(t)$ and $V(t)$ are obtained from a simplified trajectory model, and ρ_{∞} is an atmospheric density model.

Trajectory and atmosphere submodel (forcing generation). The surface heat-flux forcing $q_{\text{in}}(t)$ is generated from a point-mass reentry model with state $y = [h, v_x, v_z]^T$, where h is altitude and (v_x, v_z) are the horizontal and vertical velocity components. The governing ODEs are

$$\dot{h} = v_z, \quad (6)$$

$$\dot{v}_x = -D(h, V) \frac{v_x}{V}, \quad (7)$$

$$\dot{v}_z = -g(h) - D(h, V) \frac{v_z}{V}, \quad (8)$$

$$V = \sqrt{v_x^2 + v_z^2}, \quad D(h, V) = \frac{1}{2m} \rho_\infty(h) C_d A_{\text{ref}} V^2. \quad (9)$$

The reference area is taken as the spherical cross-section $A_{\text{ref}} = \pi R^2$. The atmosphere is modeled by an exponential density law,

$$\rho_\infty(h) = \rho_0 \exp(-h/H), \quad \rho_0 = 1.225 \text{ kg/m}^3, \quad H = 7200 \text{ m}, \quad (10)$$

and gravity varies with altitude as

$$g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \quad g_0 = 9.81 \text{ m/s}^2, \quad R_E = 6.371 \times 10^6 \text{ m}. \quad (11)$$

The trajectory ODE is integrated with a fixed step Δt_{traj} (RK4 in the implementation), and $q_{\text{in}}(t)$ is evaluated on the trajectory time grid and linearly interpolated onto the conduction time grid used by the heat solver. Figures 1–2 show the baseline altitude history (starting at 120 km and reaching ground near 210 s) and the corresponding Sutton–Graves heat-flux pulse (peaking on the order of $1 \times 10^6 \text{ W/m}^2$).

Sutton–Graves constants and geometric assumption. In this simplified spherical model, the nose radius is taken as $R_n = R$. The Sutton–Graves constant K_{SG} is held fixed across discretization studies so numerical sensitivity can be assessed consistently (see code implementation for baseline value).

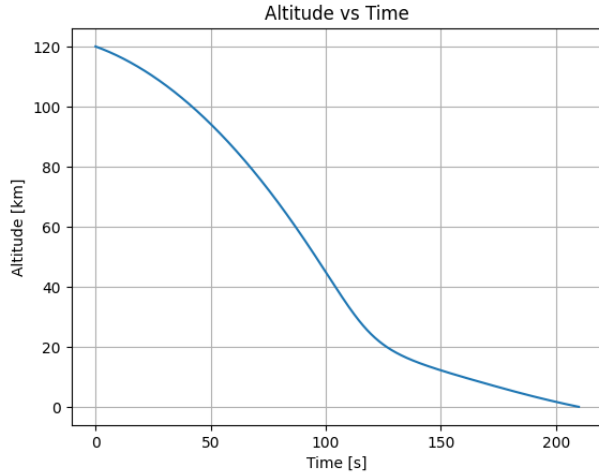


Fig. 1 Trajectory altitude history $h(t)$ used to generate the forcing.

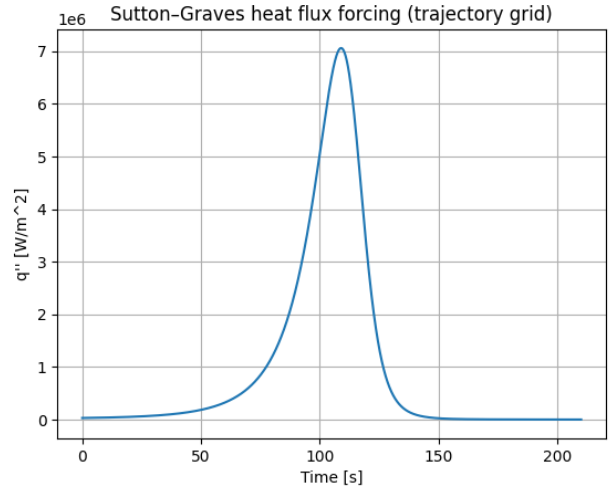


Fig. 2 Sutton–Graves inward heat-flux forcing $q_{\text{in}}(t)$ on the trajectory grid.

For a two-layer sphere with an interface at $r = r_{\text{int}}$ (TPS/structure boundary), properties are piecewise constant and we enforce standard interface conditions:

$$T(r_{\text{int}}^-, t) = T(r_{\text{int}}^+, t), \quad (12)$$

$$k^- \frac{\partial T}{\partial r} \Big|_{r_{\text{int}}^-} = k^+ \frac{\partial T}{\partial r} \Big|_{r_{\text{int}}^+}. \quad (13)$$

In the finite-volume discretization, these conditions are satisfied by construction through flux continuity at shared faces.

Questions for the dynamical system

We study how TPS material choices influence surface heating and heat penetration to the TPS back-face under reentry-like forcing. The primary quantities of interest reported in this paper are: (i) surface temperature history $T_{\text{surf}}(t) = T(R, t)$ and its peak value $T_{\text{surf,peak}}$; (ii) TPS back-face temperature $T_{\text{back}}(t) = T(r_{\text{back}}, t)$ with $r_{\text{back}} = R - \delta_{\text{TPS}}$, including $T_{\text{back,peak}}$ and the time-to-peak $t_{\text{back,peak}}$; and (iii) sensitivity of these metrics to material properties (k, ρ, c_p) while keeping the forcing model fixed. Numerical reliability is assessed through verification and discretization studies so that observed trends can be attributed to physics rather than resolution.

Mathematical derivation of dynamical system

Starting from the heat equation in divergence form without internal volumetric generation,

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T), \quad (14)$$

and imposing spherical symmetry $T = T(r, t)$ yields

$$\rho(r) c_p(r) \frac{\partial T}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(k(r) r^2 \frac{\partial T}{\partial r} \right), \quad (15)$$

which is Eq. (1). When k, ρ, c_p are constant, this reduces to the familiar expanded form

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial r^2} + \frac{2}{r} \frac{\partial T}{\partial r} \right), \quad \alpha = \frac{k}{\rho c_p}, \quad (16)$$

but our implementation retains the conservative form so that discontinuous material properties can be handled consistently.

Model parameters and study ranges

Table 2 summarizes the baseline physical and numerical parameters used in the coupled reentry-forcing simulations. Trajectory/forcing parameters are held fixed across TPS material cases unless otherwise stated; material properties are varied in Results. All numerical reference values (“ref”) correspond to the finest settings used in the discretization studies.

IV. Numerical Method

In order to solve the IBVP numerically, we discretize Eq. (1) in space using a conservative finite-volume (FV) method on concentric spherical shell control volumes, which yields an ODE system in time. This ODE system is advanced using an implicit second-order method (Crank–Nicolson / trapezoidal rule).

Justification

The reentry forcing can produce steep gradients near the surface and a stiff diffusion operator. An implicit A-stable second-order method (Crank–Nicolson) avoids restrictive explicit stability constraints while maintaining temporal accuracy. The FV discretization is well suited to spherical geometry because it enforces energy conservation over each shell volume and naturally incorporates geometric factors (areas/volumes). The resulting linear system is tridiagonal and can be solved in $O(N)$ time per step using a Thomas algorithm.

Derivation

Partition $[0, R]$ into N shells with faces at $r_{i\pm 1/2}$ and centers at r_i . For a uniform grid, $r_{i+1/2} = i\Delta r$ with $\Delta r = R/N$. Volumes and face areas are

$$V_i = \frac{4\pi}{3} (r_{i+1/2}^3 - r_{i-1/2}^3), \quad A_{i\pm 1/2} = 4\pi r_{i\pm 1/2}^2. \quad (17)$$

Integrating Eq. (1) over shell i and applying the divergence theorem gives

$$\rho_i c_{p,i} V_i \frac{dT_i}{dt} = \left[A k \frac{\partial T}{\partial r} \right]_{r_{i-1/2}}^{r_{i+1/2}}, \quad (18)$$

Table 2 Baseline parameters for coupled reentry-forcing runs and discretization studies (from the implementation notebook).

Parameter	Baseline value	Range / note
Sphere radius R	0.20 m	Fixed
TPS thickness δ_{TPS}	5×10^{-3} m	Fixed
Back-face / interface radius $r_{\text{back}} = R - \delta_{\text{TPS}}$	0.195 m	Metric location
Initial temperature T_0	300 K	Fixed
Final simulation time t_f	400 s	Coupled runs
Atmosphere $\rho_\infty(h)$	$\rho_0 e^{-h/H}$	$\rho_0 = 1.225 \text{ kg/m}^3$, $H = 7200 \text{ m}$
Gravity $g(h)$	$g_0 \left(\frac{R_E}{R_E + h} \right)^2$	$g_0 = 9.81 \text{ m/s}^2$, $R_E = 6.371 \times 10^6 \text{ m}$
Initial altitude h_0	120 km	Fixed
Initial speed V_0	8000 m/s	Fixed
Initial flight-path angle γ_0	-2°	Fixed
Vehicle mass m	200 kg	Fixed
Drag coefficient C_d	1.2	Fixed
Reference area $A_{\text{ref}} = \pi R^2$	$1.2566 \times 10^{-1} \text{ m}^2$	Fixed
Sutton–Graves nose radius R_n	$R = 0.20 \text{ m}$	Fixed
Sutton–Graves constant K_{SG}	1.0×10^{-4}	Fixed (implementation value)
TPS properties (k, ρ, c_p)	Varied	Candidate TPS sets in Results
Structure properties (k, ρ, c_p)	Fixed	Baseline structure material (see code)
Production grid N	25600	Multiple of 40 (interface-aligned)
Production Δt_{cond}	0.5 s	Fixed for Results
Production Δt_{traj}	0.05 s	Fixed for Results
Reference grid N_{ref}	51200	Reference for N and Δt_{cond} sweeps
Reference conduction step $\Delta t_{\text{cond,ref}}$	0.25 s	Reference for Δt_{cond} sweep
Reference forcing step $\Delta t_{\text{traj,ref}}$	0.02 s	Reference for Δt_{traj} sweep

Note: Resolution studies use Resin ($k = 0.20 \text{ W/(m K)}$, $\rho = 1200 \text{ kg/m}^3$, $c_p = 1200 \text{ J/(kg K)}$); structure uses ($k = 200 \text{ W/(m K)}$, $\rho = 2700 \text{ kg/m}^3$, $c_p = 900 \text{ J/(kg K)}$).

where $T_i(t)$ is the FV cell-average temperature. Face gradients are approximated by centered differences, and the face conductivity $k_{i+1/2}$ is taken as a consistent face value (harmonic mean for discontinuous k):

$$k_{i+1/2} = \frac{2k_i k_{i+1}}{k_i + k_{i+1}}. \quad (19)$$

This yields the semi-discrete system

$$M \dot{\mathbf{T}}(t) = K \mathbf{T}(t) + \mathbf{f}(t), \quad (20)$$

where $M = \text{diag}(\rho_i c_{p,i} V_i)$ and K is tridiagonal. The forcing $\mathbf{f}(t)$ includes the prescribed surface heat flux (Neumann condition) and, in verification cases, any manufactured volumetric source.

At the center, the inner face area satisfies $A_{1/2} = 0$, so the symmetry condition is enforced automatically. At the surface, the prescribed inward flux $q_{\text{in}}(t)$ contributes net power $q_{\text{in}}(t)A(R)$ into the outermost control volume, consistent with Eq. (4).

Time integration of Eq. (20) with the trapezoidal rule gives

$$\left(M - \frac{\Delta t}{2} K \right) \mathbf{T}^{n+1} = \left(M + \frac{\Delta t}{2} K \right) \mathbf{T}^n + \Delta t \mathbf{f}^{n+1/2}, \quad (21)$$

where $\mathbf{f}^{n+1/2}$ is evaluated consistently at the time midpoint (or by averaging endpoints), producing a tridiagonal solve each step.

Algorithmic summary (solver workflow).

Given R , layer definitions, initial temperature T_0 , and discretization (N , Δt_{cond} , Δt_{traj}):

- 1) Build the uniform radial FV grid and compute $\{V_i, A_{i\pm 1/2}\}$. Assign piecewise-constant properties and assemble M and tridiagonal K using harmonic face conductivities.
- 2) Integrate the trajectory ODEs on the Δt_{traj} grid, compute $q_{\text{in}}(t)$ via Sutton–Graves, and interpolate q_{in} onto the conduction times t^n .
- 3) For each conduction step, apply Crank–Nicolson to Eq. (20) and solve the resulting tridiagonal system for $\mathbf{T}^{n+1}$ using a Thomas algorithm.
- 4) Post-process $T_{\text{surf}}(t)$ and $T_{\text{back}}(t)$ (interpolated to r_{back}) and compute the metrics used in Results.

V. Implementation

The numerical method is implemented in modular form to mirror the derivation above:

- **FV operator build:** Construct uniform radial grid, compute face areas and shell volumes, assign piecewise-constant material properties by layer, and assemble diagonal M and tridiagonal K for spherical FV conduction.
- **Implicit CN time step:** Each time step solves a tridiagonal linear system using a Thomas algorithm. The solve cost scales linearly with N .
- **Boundary-flux forcing:** A positive inward surface flux $q_{\text{in}}(t)$ injects power $q_{\text{in}}A(R)$ into the outermost control volume. In the reentry case, $q_{\text{in}}(t)$ is computed from Sutton–Graves along a trajectory time grid and then interpolated onto the conduction time grid.
- **Quantity extraction:** FV unknowns are cell averages; temperatures at specific radii (e.g., TPS back-face) are obtained by linear interpolation between adjacent cell centers.

Numerical reliability. Correctness and consistency with the course emphasis on verification are demonstrated via: (i) a method of manufactured solutions (MMS) study confirming second-order convergence in space and time under refinement, and (ii) an analytical cross-check for a homogeneous sphere with constant inward surface heat flux using a cell-averaged analytical reference consistent with FV unknowns.

Reproducibility All results in this paper are reproducible from the accompanying repository: the production solver is separated from verification tests (MMS and analytical cross-check) and from discretization/trajectory studies used to generate the tables and figures. The repository includes scripts/notebooks that regenerate Tables 3–6 and Figures 1–9 from scratch. Code and run instructions are available at: github.com/Landon-Lopez57/AE370-Project-2.

Verification

In this paper, we use verification to denote checks of numerical correctness (error convergence and analytical cross-checks). In addition to verification, we include discretization parameter studies for the coupled reentry-forcing problem to select simulation parameters that are accurate enough for the engineering metrics reported in Results.

1. Method of Manufactured Solutions (MMS)

We verify the FV spatial discretization and the Crank–Nicolson (CN) time integrator using a smooth manufactured solution that is regular at the origin and satisfies a homogeneous Neumann condition at the surface:

$$T_{\text{ex}}(r, t) = T_{\text{ref}} + A e^{-t/\tau} j_0(\lambda r), \quad j_0(x) = \frac{\sin x}{x}. \quad (22)$$

The parameter λ is chosen so that $\partial T_{\text{ex}}/\partial r(R, t) = 0$, which for j_0 yields

$$\tan(x) = x, \quad x = \lambda R. \quad (23)$$

To make T_{ex} an exact solution of the forced PDE, we add a manufactured volumetric source $\dot{q}_{\text{ex}}(r, t)$ such that

$$\rho c_p \frac{\partial T_{\text{ex}}}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(k r^2 \frac{\partial T_{\text{ex}}}{\partial r} \right) + \dot{q}_{\text{ex}}(r, t). \quad (24)$$

Because FV unknowns represent cell averages, the analytical reference is compared in a cell-averaged sense:

$$\bar{T}_{\text{ex},i}(t) = \frac{1}{V_i} \int_{r_{i-1/2}}^{r_{i+1/2}} T_{\text{ex}}(r, t) 4\pi r^2 dr,$$

evaluated numerically using a fine quadrature within each cell.

Error metric. We measure error at $t = t_f$ using a volume-weighted discrete L_2 norm,

$$\|e\|_{2,V} = \left(\frac{\sum_{i=0}^{N-1} V_i |T_i(t_f) - \bar{T}_{\text{ex},i}(t_f)|^2}{\sum_{i=0}^{N-1} V_i} \right)^{1/2}. \quad (25)$$

Spatial refinement (FV). With a very small time step ($\Delta t = 1 \times 10^{-3}$ s) and an oscillatory mode ($n = 8$) to expose spatial error, we refine $N \rightarrow 2N$ and estimate the observed order

$$p_r \approx \frac{\log(\|e_N\|/\|e_{2N}\|)}{\log(2)}. \quad (26)$$

Table 3 shows $p_r \approx 2$, confirming second-order spatial accuracy.

N	$\Delta r = R/N$ [m]	$\ e\ _{2,V}$ [K]	p_r
30	6.667×10^{-3}	6.549×10^{-3}	—
60	3.333×10^{-3}	1.652×10^{-3}	1.987
120	1.667×10^{-3}	4.140×10^{-4}	1.997
240	8.333×10^{-4}	1.036×10^{-4}	1.999
480	4.167×10^{-4}	2.591×10^{-5}	1.999
960	2.083×10^{-4}	6.488×10^{-6}	1.998

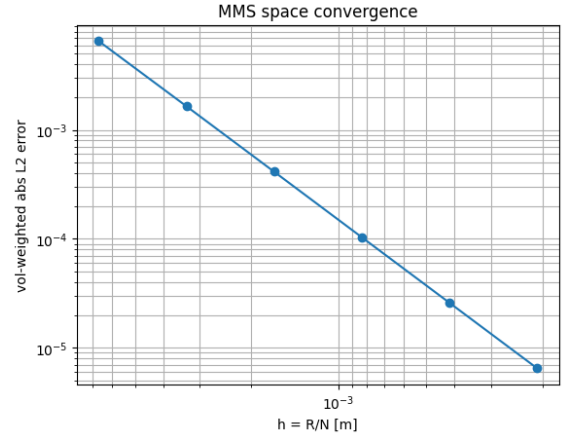


Table 3 MMS spatial convergence (volume-weighted absolute L_2 error) using an oscillatory mode ($n = 8$) and $\Delta t = 1 \times 10^{-3}$ s ($t_f = 0.5$ s).

Fig. 3 MMS spatial convergence: $\|e\|_{2,V}$ vs. Δr (log-log).

Temporal refinement (CN). With a fine grid ($N = 2000$) and a smoother mode ($n = 2$), we refine $\Delta t \rightarrow \Delta t/2$ and estimate

$$p_t \approx \frac{\log(\|e_{\Delta t}\|/\|e_{\Delta t/2}\|)}{\log(2)}. \quad (27)$$

Table 4 shows near second-order temporal accuracy for CN.

Δt [s]	$\ e\ _{2,V}$ [K]	p_t
1.00×10^{-1}	5.158×10^{-4}	–
5.00×10^{-2}	1.293×10^{-4}	1.995
2.50×10^{-2}	3.273×10^{-5}	1.983
1.25×10^{-2}	8.577×10^{-6}	1.932

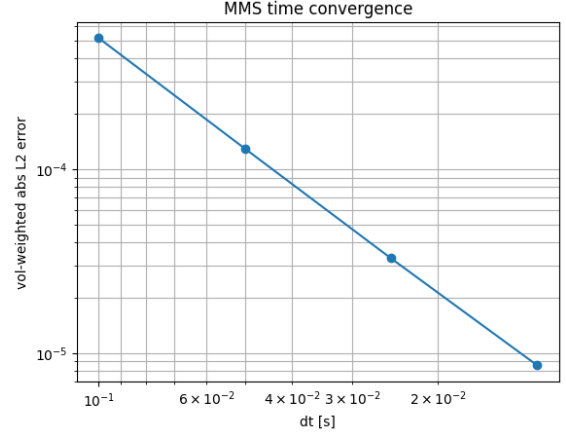


Table 4 MMS temporal convergence. Volume-weighted absolute L_2 error with $N = 2000$ and $t_f = 0.5$ s.

Fig. 4 MMS temporal convergence: $\|e\|_{2,V}$ vs. Δt (log-log).

2. Analytical cross-check: constant-flux homogeneous sphere

We perform an analytical cross-check using a homogeneous sphere with constant properties (ρ, c_p, k) , uniform initial temperature $T(r, 0) = T_0$, and constant inward surface heat flux q_{in} at $r = R$. This isolates spherical geometry factors, flux forcing, operator assembly, and CN stepping without any trajectory coupling.

The analytical solution is implemented as a particular + homogeneous decomposition

$$T(r, t) = T_p(r, t) + \theta(r, t), \quad (28)$$

with the particular solution

$$T_p(r, t) = T_0 + Br^2 + 6\alpha Bt, \quad \alpha = \frac{k}{\rho c_p}, \quad B = \frac{q_{in}}{2kR}, \quad (29)$$

which satisfies the inward-flux boundary condition $kT_r(R, t) = q_{in}$. The remaining transient $\theta(r, t)$ satisfies homogeneous Neumann conditions and is expanded in spherical eigenmodes:

$$\theta(r, t) = c_0 + \sum_{n=1}^{N_{terms}} c_n j_0(\lambda_n r) e^{-\alpha \lambda_n^2 t}, \quad \tan(x) = x, \quad x = \lambda R. \quad (30)$$

Cell-averaged comparison (consistent with FV unknowns). Because FV unknowns represent shell averages, we compute the analytical reference as

$$\bar{T}_i(t) = \frac{1}{V_i} \int_{r_{i-1/2}}^{r_{i+1/2}} T(r, t) 4\pi r^2 dr, \quad (31)$$

evaluated numerically within each cell using fine quadrature.

Numerical run and error metrics. For $N = 960$, $\Delta t = 0.01$ s, and a 200-term truncation, the observed errors at $t = t_f$ are

$$\|e\|_{2,V} = 3.40 \times 10^{-3} \text{ K}, \quad (32)$$

$$\|e\|_{\infty} = 1.61 \times 10^{-2} \text{ K}. \quad (33)$$

Agreement at the $O(10^{-2})$ K level provides strong evidence that the spherical FV operator, the inward-flux forcing, and CN stepping are implemented correctly.

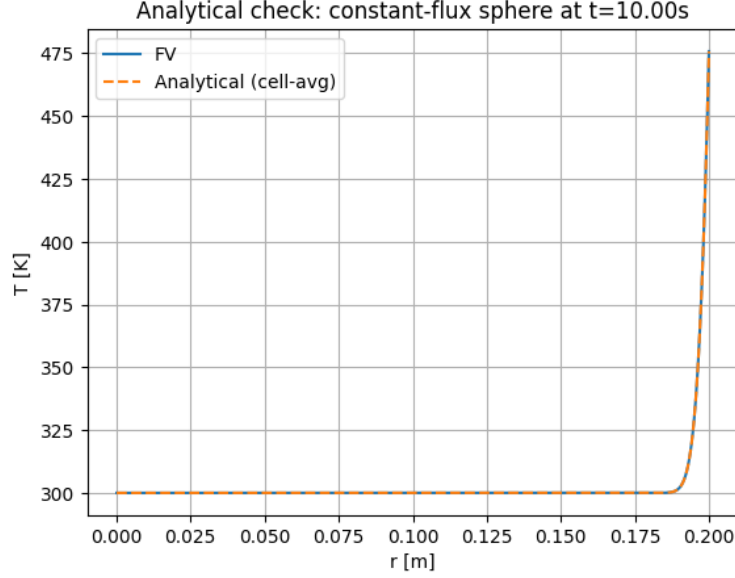


Fig. 5 A2 analytical check for a homogeneous sphere with constant inward surface heat flux. FV cell averages are compared to cell-averaged analytical values at $t = t_f$ ($N = 960$, $\Delta t = 0.01$ s, 200-term truncation).

Simulation Parameters

The coupled reentry simulations contain two independent discretization components:

- **Conduction discretization:** $(N, \Delta t_{\text{cond}})$ for the FV–CN heat solver.
- **Forcing sampling:** Δt_{traj} for trajectory integration and heat-flux sampling, followed by interpolation of $q_{\text{in}}(t)$ onto the conduction time grid.

This section is *not* verification; instead, it is a parameter-selection study to identify $(N, \Delta t_{\text{cond}}, \Delta t_{\text{traj}})$ that is accurate enough for the engineering metrics used in Results while remaining computationally practical.

Metrics and acceptance criterion. For a metric location $x \in \{\text{surf}, \text{back}\}$, define

$$T_x(t) = \begin{cases} T(R, t), & x = \text{surf}, \\ T(r_{\text{back}}, t), & x = \text{back}, \end{cases} \quad T_{x,\text{peak}} = \max_{t \in [0, t_f]} T_x(t).$$

We report (i) the relative peak error

$$e_{\text{peak},x} = \frac{|T_{x,\text{peak}} - T_{x,\text{peak}}^{\text{ref}}|}{|T_{x,\text{peak}}^{\text{ref}}|}, \quad (34)$$

and (ii) the maximum-in-time absolute error

$$e_{\infty,x} = \max_{t \in [0, t_f]} |T_x(t) - T_x^{\text{ref}}(t)|. \quad (35)$$

Because the surface temperature undergoes very large excursions under reentry heating, an absolute $e_{\infty,\text{surf}}$ can appear large even when the *relative* peak metrics are already well converged. Since the primary engineering quantity of interest is the TPS back-face response, we prioritize $e_{\text{peak},\text{back}}$ (and the back-face trace) when selecting N and the time steps.

Back-face location and interface alignment. We define the TPS back-face at

$$r_{\text{back}} = R - \delta_{\text{TPS}}, \quad \delta_{\text{TPS}} = 0.005 \text{ m}, \quad R = 0.20 \text{ m} \Rightarrow r_{\text{back}} = 0.195 \text{ m}.$$

To avoid smearing the TPS/structure interface, we select N so the interface aligns with a grid face. With $\Delta r = R/N$,

$$\frac{\delta_{\text{TPS}}}{\Delta r} = \frac{N}{40} \in \mathbb{Z} \Rightarrow \text{choose } N \text{ as a multiple of 40}.$$

Conduction time-step selection: Δt_{cond} sweep (fixed spatial grid). We first assess time-integration error in the heat solver by sweeping Δt_{cond} while holding the forcing sampling fixed at $\Delta t_{\text{traj}} = 0.05 \text{ s}$. To reduce contamination from spatial error, this sweep is performed on a fine grid ($N = 51200$), and the finest time step in the sweep ($\Delta t_{\text{cond}} = 0.25 \text{ s}$) is used as the reference solution. For Δt_{cond} , we apply a strict absolute criterion on the back-face trace,

$$e_{\infty, \text{back}} < 0.02 \text{ K},$$

since this directly bounds the maximum temperature history discrepancy at the TPS back-face. Table 5 shows that $\Delta t_{\text{cond}} = 0.5 \text{ s}$ satisfies this criterion, while $\Delta t_{\text{cond}} = 1.0 \text{ s}$ does not. We therefore choose

$$\Delta t_{\text{cond}} = 0.5 \text{ s}.$$

Table 5 Conduction time-step sweep at fixed $N = 51200$ (forcing sampling fixed at $\Delta t_{\text{traj}} = 0.05 \text{ s}$). **Reference:** $\Delta t_{\text{cond}} = 0.25 \text{ s}$. **Acceptance criterion:** $e_{\infty, \text{back}} < 0.02 \text{ K}$.

$\Delta t_{\text{cond}} [\text{s}]$	$e_{\text{peak}, \text{back}}$	$e_{\infty, \text{back}} [\text{K}]$	Meets $e_{\infty, \text{back}} < 0.02 \text{ K}$
1.0	6.0×10^{-6}	0.044084	No
0.5	1.0×10^{-6}	0.008828	Yes
0.25	0	0	Yes (reference)

Spatial resolution selection: N sweep (fixed Δt_{cond}). We next sweep the number of radial control volumes N while holding $\Delta t_{\text{traj}} = 0.05 \text{ s}$ fixed and using a small conduction time step ($\Delta t_{\text{cond}} = 0.25 \text{ s}$) so that time-discretization effects are negligible during the spatial study. The finest grid in the sweep ($N = 51200$) is used as the reference. In principle, one could enforce the strict absolute criterion $e_{\infty, \text{back}} < 0.02 \text{ K}$ for the spatial selection as well; however, Table 6 shows that this would require substantially finer grids than are computationally practical for routine production runs. Instead, we use the peak-based engineering criterion

$$e_{\text{peak}, \text{back}} < 1\%,$$

which targets the back-face peak temperature used in Results. Table 6 shows that $N = 25600$ is the smallest grid meeting the 1% criterion, while also satisfying the interface-alignment constraint (N multiple of 40). We therefore choose

$$N = 25600.$$

Table 6 Spatial resolution sweep at fixed $\Delta t_{\text{cond}} = 0.25 \text{ s}$ (forcing sampling fixed at $\Delta t_{\text{traj}} = 0.05 \text{ s}$). **Reference:** $N = 51200$. **Acceptance criterion:** $e_{\text{peak}, \text{back}} < 1\%$.

N	$\Delta r = R/N [\text{m}]$	$e_{\text{peak}, \text{back}}$	$e_{\infty, \text{back}} [\text{K}]$	Meets $e_{\text{peak}, \text{back}} < 1\%$
12800	1.563×10^{-5}	0.029886	28.369990	No
25600	7.813×10^{-6}	0.009962	9.456577	Yes
51200	3.906×10^{-6}	0	0	Yes (reference)

Forcing sampling sensitivity: Δt_{traj} sweep. Finally, we evaluate sensitivity to the trajectory/forcing sampling step Δt_{traj} , which controls the resolution of the Sutton–Graves heat-flux pulse before it is interpolated onto the conduction grid. In this sweep, we hold the conduction discretization fixed ($N = 6400$, $\Delta t_{\text{cond}} = 0.25$ s) and compare against a fine forcing reference $\Delta t_{\text{traj}} = 0.02$ s. Table 7 shows that $\Delta t_{\text{traj}} \leq 0.10$ s yields essentially converged back-face peaks (and extremely small $e_{\infty, \text{back}}$). We therefore choose

$$\Delta t_{\text{traj}} = 0.05 \text{ s}$$

to make forcing discretization error negligible.

Table 7 Forcing sampling sensitivity at fixed $N = 6400$ and $\Delta t_{\text{cond}} = 0.25$ s. Reference: $\Delta t_{\text{traj}} = 0.02$ s.

Δt_{traj} [s]	$T_{\text{back, peak}}$ [K]	$e_{\text{peak, back}}$	$e_{\infty, \text{back}}$ [K]
0.50	941.254039	1.196161×10^{-3}	1.124546
0.25	941.254799	1.196968×10^{-3}	1.125305
0.10	940.129464	3.117896×10^{-8}	0.000158
0.05	940.129495	1.299780×10^{-9}	0.000007
0.02	940.129493	0	0

Final recommended discretization (used in Results). Combining the above studies, all reported production simulations use:

$$\Delta t_{\text{traj}} = 0.05 \text{ s}, \quad \Delta t_{\text{cond}} = 0.5 \text{ s}, \quad N = 25600.$$

VI. Results and Discussion

This section presents results that address the primary project question: how do TPS material choices influence (i) surface heating and (ii) heat penetration to the TPS back face under a reentry like, trajectory generated heat-flux forcing. Rather than showing temperature profiles at many arbitrary times, results are summarized using engineering-relevant metrics: the surface temperature history $T_{\text{surf}}(t) = T(R, t)$ and its peak $T_{\text{surf, peak}}$, and the TPS back face temperature $T_{\text{back}}(t) = T(r_{\text{back}}, t)$ with peak $T_{\text{back, peak}}$ and time to peak $t_{\text{back, peak}}$. All “production” results in this section use the discretization selected in Sec. V: $\Delta t_{\text{traj}} = 0.05$ s, $\Delta t_{\text{cond}} = 0.5$ s, and $N = 25600$.

A. Baseline coupled reentry heating response

Figure 6 summarizes the coupled baseline run by plotting the imposed surface heat flux $q_{\text{in}}(t)$ alongside the resulting surface and back face temperatures. The heating pulse generated by the trajectory/atmosphere model produces a rapid surface temperature rise, while the back face response is delayed and smoothed by diffusion through the TPS. This phase lag is a key feature for TPS design: peak structural risk (as measured by T_{back}) generally occurs after the peak of the aerodynamic heating input.

For the baseline configuration, the back face peak temperature is approximately $T_{\text{back, peak}} \approx 35255.11$ K, with negligible sensitivity to forcing sampling for $\Delta t_{\text{traj}} \leq 0.10$ s. The time of peak $t_{\text{back, peak}}$ is reported from the temperature history and is used as a secondary indicator of how quickly heat penetrates the TPS.

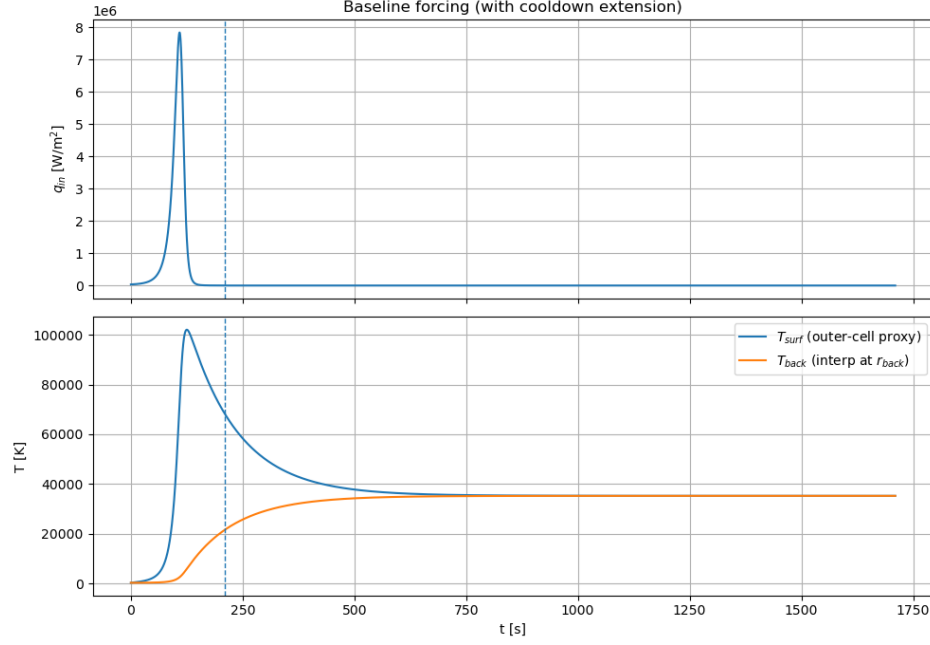


Fig. 6 Baseline coupled run: imposed inward surface heat flux $q_{in}(t)$ (top) and resulting temperature metrics (bottom): surface temperature $T_{surf}(t)$ and TPS back-face temperature $T_{back}(t)$ at $r_{back} = R - \delta_{TPS}$. Vertical markers (optional) can indicate the times of peak heat flux, peak surface temperature, and peak back-face temperature.

B. Interpreting penetration: temperature distribution in radius and time

To visualize how the heating pulse propagates inward, Figure 7 shows a compact spacetime view of the temperature field. Early in the trajectory, heating is confined to a thin near surface thermal layer. As time progresses, the thermal front advances inward, eventually raising the TPS back face temperature. This representation directly connects the forcing time scale to the diffusive penetration time scale and helps explain trends observed in the parameter studies below.

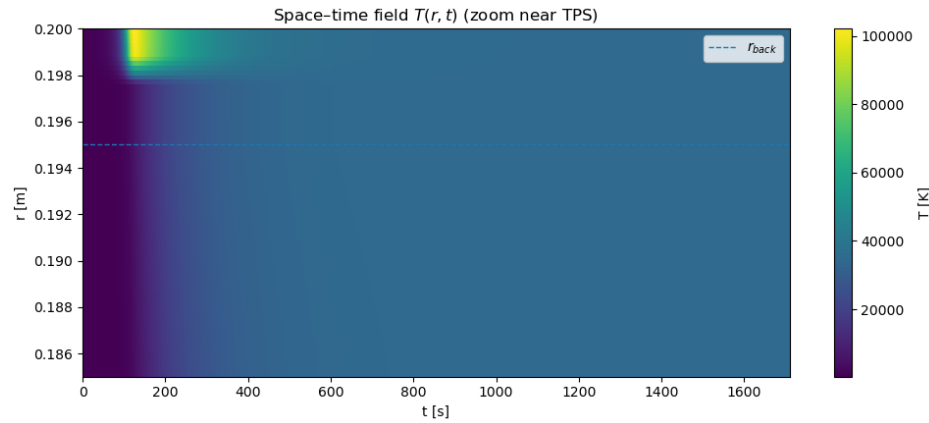


Fig. 7 Baseline run temperature field visualized as a space–time contour $T(r, t)$ (restricted to the outer region or the full radius). Overlaying the TPS/structure interface location r_{back} highlights when significant heating reaches the back-face.

C. TPS material sensitivity study

The remaining results quantify how varying TPS properties changes (i) peak temperatures and (ii) time-to-peak. Each sweep varies one TPS parameter at a time while holding the forcing model fixed and keeping the inner structure properties unchanged. Results are summarized primarily using $(T_{\text{surf,peak}}, T_{\text{back,peak}}, t_{\text{back,peak}})$.

1. Effect of TPS conductivity k_{TPS}

Figure 8 reports how varying TPS thermal conductivity changes the surface peak and back face peak. Increasing k_{TPS} reduces thermal resistance, promoting faster inward heat transport. Consequently, $T_{\text{back,peak}}$ increases and occurs earlier (smaller $t_{\text{back,peak}}$), while the surface peak $T_{\text{surf,peak}}$ tends to decrease because the surface gradient required to conduct the imposed heat flux is reduced. Conversely, decreasing k_{TPS} increases insulation: it raises the surface peak but lowers and delays the back face peak.

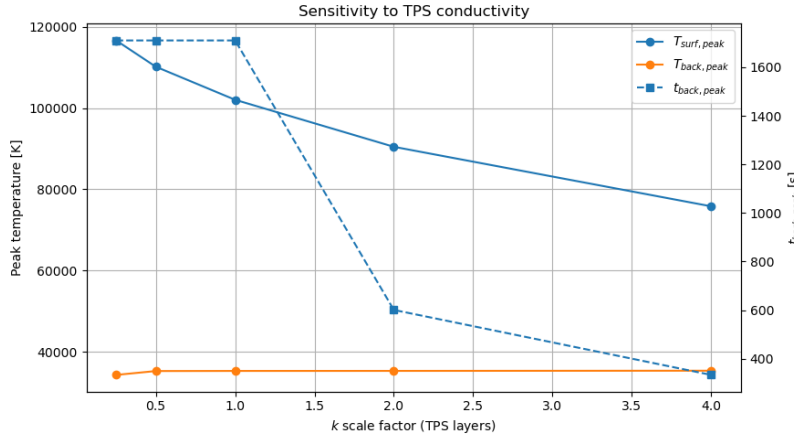


Fig. 8 Sensitivity to TPS conductivity k_{TPS} : variation of $T_{\text{surf,peak}}$, $T_{\text{back,peak}}$, and (optional) $t_{\text{back,peak}}$ versus k_{TPS} .

2. Effect of volumetric heat capacity $\rho_{\text{TPS}}c_{p,\text{TPS}}$

Figure 9 shows sensitivity to thermal mass through ρ_{TPS} and/or $c_{p,\text{TPS}}$. Increasing ρc_p increases the energy required to raise TPS temperature, reducing both $T_{\text{surf,peak}}$ and $T_{\text{back,peak}}$ and typically delaying the back face peak (larger $t_{\text{back,peak}}$). Decreasing ρc_p produces the opposite trend: faster warming and higher peaks.

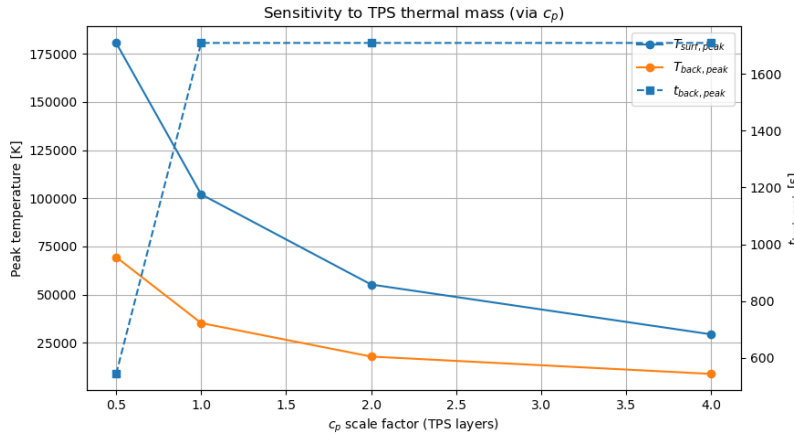


Fig. 9 Sensitivity to TPS thermal mass (via ρ_{TPS} and/or $c_{p,\text{TPS}}$): variation of $T_{\text{surf,peak}}$, $T_{\text{back,peak}}$, and $t_{\text{back,peak}}$.

D. Summary of trends and design implications

Table 8 summarizes the qualitative sensitivity of peak temperatures and time-to-peak to key TPS parameters. The main trade is between surface protection and internal protection: reducing conductivity (or diffusivity) tends to increase surface temperature while improving insulation at the TPS back-face, whereas increasing conductivity reduces surface temperature but increases and accelerates back-face heating. Increasing thermal mass reduces temperatures throughout and tends to delay the back-face peak.

Table 8 Qualitative sensitivity of key metrics to TPS parameters (holding forcing fixed).

Parameter increase	$T_{\text{surf,peak}}$	$T_{\text{back,peak}}$	$t_{\text{back,peak}}$
$k_{\text{TPS}} \uparrow$	$\downarrow$	$\uparrow$	$\downarrow$
$(\rho c_p)_{\text{TPS}} \uparrow$	$\downarrow$	$\downarrow$	$\uparrow$
$\alpha_{\text{TPS}} \uparrow$	(weak)	$\uparrow$	$\downarrow$

E. Note on surface-temperature extraction (FV resolution sensitivity)

The discretization studies show that the back face temperature history is tightly converged under the selected production setup, while surface temperature measures are substantially more sensitive to spatial resolution and time step due to steep near-surface gradients under imposed heat flux. Therefore, surface temperature results are interpreted primarily in terms of robust trends across parameter variations, and back-face metrics are treated as the primary engineering outputs for TPS performance in this study. (If desired, a higher-resolution reconstruction of $T(R, t)$ from FV cell averages and the imposed Neumann boundary gradient can be used to further reduce surface sensitivity.)

VII. Team Roles and Contributions

(Danfeng Ye) I led the Verification and Simulation Parameters sections and made major improvements to the solver workflow. I implemented the spherical FV–CN conduction solver with tridiagonal solves and the inward heat-flux boundary condition, and coupled the reentry forcing by sampling $q_{\text{in}}(t)$ on a trajectory grid and interpolating it onto the conduction grid. I built and ran the verification suite (MMS second-order space/time convergence and an analytical constant-flux sphere cross-check using FV-consistent cell-averaged references), and completed the N , Δt_{cond} , and Δt_{traj} sensitivity studies to select production settings and finalize the run scripts for the reported figures and tables.

(Jeremy Zang) I was responsible for writing various sections of the report. Specifically, I authored the initial drafts of the introduction and dynamical system sections, and was responsible for editing the numerical methods section.

(Landon Lopez) I was only responsible for the initial structure of the implementation and the reentry dynamics. I also managed the github repository.

(Maxwell Lutter) I was responsible for writing parts of the validation and numerical reliability section and the implementation section. I ensured that the methods and derivations in these sections were consistent with our code and with the course material. I also helped with some interpretation of the validation results to work through any errors or inconsistencies, and helped with plotting and representing the data for these sections. Furthermore I helped with report structure and some initial drafts and edits of the report.

(Kenny Carter) I was responsible for completing the results and discussion section. I generated baseline reentry heating plots and used them to define the key performance metrics that we outlined. I also created a space time visualization to illustrate the thermal penetration and demonstrate the diffusion behavior.

Acknowledgments

This work was completed as part of AE370 at the University of Illinois Urbana–Champaign. Generative AI (ChatGPT) was used as a writing aid to help edit wording, improve document organization, and draft LaTeX-ready descriptions of methods and verification procedures. All governing equations, numerical implementation, verification tests, parameter studies, simulations, and plots were produced and checked by the authors. The authors are responsible for the technical correctness of all reported results.

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